6. Experimental Results and Analysis

6.1 Experimental Settings

The rice leaf task contains five classes: Brown Spot, Leaf Scald, Rice Blast, Rice Tungro and Sheath Blight. The available archive provides 1,106 labelled thumbnail images and five pre-generated splits using seeds 11, 22, 33, 44 and 55. Each split assigns approximately 70% of samples to training, 15% to validation and 15% to testing. Performance is reported for white-background images, field-background images and both backgrounds together.

The experiment reused Role 2's feature extractor: resizing to 256 × 256 pixels, 3 × 3 Gaussian smoothing, a 16 × 4 × 4-bin HSV histogram and uniform local binary pattern histograms at (8, 1) and (16, 2). A StandardScaler was fitted to training features only. Classification used an RBF-kernel SVM with C = 1, gamma = scale and balanced class weights.

6.2 Results

Table 1 reports the mean and sample standard deviation over five independently shuffled test splits. Macro metrics give each disease category equal weight, reducing the risk that a more frequent class dominates the assessment.

Background	Accuracy	Macro precision	Macro recall	Macro F1
White	48.95 ± 2.71	51.13 ± 4.14	45.24 ± 2.54	46.19 ± 2.72
Field	46.40 ± 7.13	48.12 ± 8.20	45.99 ± 7.51	45.17 ± 6.82
Combined	48.17 ± 2.93	48.47 ± 3.04	46.46 ± 3.49	46.66 ± 3.25

Table 1. Measured thumbnail SVM results (%), mean ± sample SD over five runs.

Across combined-background tests, mean accuracy was **48.17%** and macro F1 was **46.66%**. White-background accuracy averaged **48.95%**, compared with **46.40%** for field backgrounds. The standard deviation was larger for field images (7.13 percentage points) than white images (2.71 percentage points).

6.3 Discussion

The modest classification performance suggests that colour and local texture alone do not consistently distinguish the five disease categories at thumbnail resolution. The field-background subset was slightly harder on average, plausibly because foliage, soil and uneven lighting introduce visual information unrelated to symptoms. This is an interpretation rather than proof of the cause. Differences in the number and difficulty of images within each split may also contribute.